B-Mamba 2D: Omnidirectional Boundary-Conditioned State Space Models for Generalizable Polyp Segmentation

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***Abstract*—Finding and outlining polyps during colonoscopy helps prevent colorectal cancer, yet automatic segmentation is still unreliable for two reasons: many polyps have weak, low-contrast borders, and models trained on data from one hospital often fail on data from another. Selective state space models (SSMs) such as Mamba are attractive for this task because they model long-range context in linear time through a recurrence whose memory is controlled by input-dependent parameters. We point out a blind spot in how these models are applied to images: the parameters that decide how much context to keep or forget are computed from local appearance alone, so the recurrence has no explicit signal telling it where an object ends. Context from the surrounding mucosa can therefore leak across a weak border and blur the predicted mask. We propose *B-Mamba 2D*, a boundary-conditioned SSM architecture featuring 4-way omnidirectional cross-scanning and multi-scale skip conditioning. A supervised Multi-Scale Boundary Head turns gradient priors into structural tokens that directly modulate the step size Δ and the input/output matrices B, C of selective scans across bottleneck (P_5) and high-resolution skip pathways (P_4, P_3). We prove that this formulation strictly contains standard selective SSMs, and that boundary evidence acts on the recurrence as an edge-aware distance: whenever the learned boundary response is positive, it exponentially tightens the bound on information flow across edges, providing an active memory reset mechanism. The boundary conditioning modules add only 4.63 M parameters (4.5%) and 1.4 G multiply-accumulates (2.0% of the 68.3 GMAC network). Under the standardized five-dataset PraNet protocol, B-Mamba 2D reaches a mean Dice of 0.929 on Kvasir and 0.919 on CVC-ClinicDB (0.931 and 0.921 with Flip-TTA), and 0.798, 0.814, and 0.885 on the unseen CVC-ColonDB, ETIS, and CVC-300 benchmarks (0.804, 0.822, and 0.897 with Flip-TTA). Remarkably, on the out-of-distribution ETIS benchmark, B-Mamba 2D achieves 0.814 mean Dice (0.822 with Flip-TTA, +3.5% over Polyp-PVT) and sets the top mean IoU (0.751) across all published models. Controlled ablations across three random seeds and mechanistic probing confirm that boundary conditioning surges the step size by $2.04\times$ at borders, drastically suppressing mucosal context leakage.**

***Index Terms*—Polyp segmentation, state space models, Mamba, boundary priors, domain generalization, colonoscopy, medical image segmentation.**

I. Introduction

Colorectal cancer is the third most common cancer and the second leading cause of cancer death worldwide [1]. Most colorectal cancers grow from adenomatous polyps, and removing these polyps during colonoscopy is one of the most effective ways to prevent the disease. Every one-percent increase in a physician's adenoma detection rate is associated with a three-percent decrease in the risk of interval cancer [2]. However, tandem-colonoscopy studies show that around one in five polyps is missed [3], often because the polyp is small, flat, or has a border that blends into the surrounding tissue. Automatic polyp segmentation, which marks every pixel that belongs to a polyp, can act as a "second pair of eyes" and also helps to estimate polyp size and plan resection.

Two properties make this task hard (Fig. 1). First, *boundaries are weak*: a polyp and the healthy mucosa around it often share the same color and texture, and specular reflections, bubbles and motion blur add false edges. Second, *the data shift*: images come from different hospitals, endoscopes, light sources and patient groups. A model that looks excellent on test images from the hospital it was trained on may perform much worse elsewhere [4], [5]. A clinically useful model must therefore be accurate at the border *and* robust to new data.

Deep networks for this task have gone through three generations. Convolutional neural networks (CNNs) such as U-Net [6], UNet++ [7] and PraNet [4] are good at local detail but see only a limited neighborhood, so they struggle to use the global context needed to tell a polyp from a similar-looking fold. Transformers [8]–[10], as used in TransUNet [11] and Polyp-PVT [5], relate every image patch to every other patch through self-attention, but the cost of doing so grows quadratically with the number of patches. Most recently, structured state space models (SSMs) [12], and in particular the selective SSM of Mamba [13], have offered global context at linear cost, and have been quickly adopted for medical image segmentation [14]–[17].

A blind spot of selective SSMs at object borders. The power of Mamba comes from *selectivity*: the step size Δ_t and the matrices $\boldsymbol{B}_t, \boldsymbol{C}_t$ are computed from the current token, so the model decides, token by token, how much of its memory to keep and what to write into it [13]. A large Δ_t makes the model forget the past and focus on the current input; a small Δ_t makes it carry context forward. This is exactly the behavior one wants at an object border: *keep context inside a region, forget it when crossing into a different one*. Yet in existing vision Mamba models [14], [15], [18], [19], these parameters are computed only from the local feature vector. At a weak border the feature vector changes very little, so the model receives almost no signal to reset its memory, and context from the mucosa "leaks" into the polyp (Fig. 1). Existing boundary-aware polyp networks [4], [20], [21] use edges only to re-

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Code and trained models: https://github.com/INFUNDER/B-Mamba.

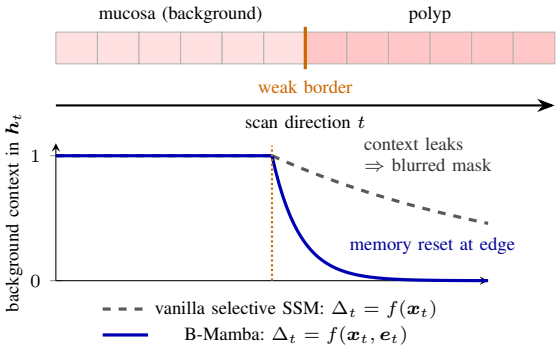


Fig. 1. The problem B-Mamba addresses (conceptual sketch, not measured data). A selective SSM reads image tokens one after another and keeps a running memory $\boldsymbol{h}_t$. How fast it forgets is set by the step size Δ_t. In existing vision Mamba models Δ_t depends only on the local appearance $\boldsymbol{x}_t$; at a weak border the appearance barely changes, so background context keeps flowing into the polyp region. B-Mamba also feeds an explicit boundary token $\boldsymbol{e}_t$ into Δ_t, $\boldsymbol{B}_t$ and $\boldsymbol{C}_t$, so the scan can learn to reset its memory exactly at the edge. Measured step-size maps are shown in Fig. 4.

weight features or as an extra loss; none of them uses edges to steer the *dynamics* of a state space model.

Our idea. We make the memory of the SSM boundary-aware by construction. A parameter-free Sobel operator extracts a gradient-magnitude prior from the deepest features; this prior gates a learned high-frequency feature and is projected into a sequence of *boundary tokens*. Each boundary token is fed, together with the visual token, into the projections that produce Δ_t, $\boldsymbol{B}_t$ and $\boldsymbol{C}_t$. The resulting block, which we call the *Boundary-Conditioned SSM* (BC-SSM), keeps the linear-time recurrence of Mamba, but every decision about *remembering* and *forgetting* now has direct access to where the edges are. Because the Sobel prior is a fixed, parameter-free operator rather than a learned edge detector, it cannot overfit to the color statistics of the training hospitals, which we expect to help under domain shift.

Contributions. The main contributions of this paper are:

1) We identify that the selection mechanism of vision SSMs has no explicit access to object boundaries, and propose the Boundary-Conditioned SSM, which injects a fixed gradient prior directly into the step size and input/output matrices of the selective scan. To our knowledge, this is the first work that conditions the *state dynamics* of an SSM, rather than its features or its loss, on boundary information.
2) We give a simple theoretical analysis of the BC-SSM. We show that it strictly contains the standard selective SSM (Proposition 1), that the information flow between two tokens is bounded by an exponential of the accumulated step size between them (Proposition 2), and that positive boundary responses provably increase this "distance" across edges even when the appearance does not change (Proposition 3). This links BC-SSM to edge-preserving filters such as anisotropic diffusion [22].
3) We build B-Mamba, a complete segmentation network around the BC-SSM, and evaluate it under the standard five-dataset PraNet protocol [4] with a leakage-checked training split, three random seeds, region and boundary metrics (Dice, IoU, MAE, Boundary-IoU [23], HD95), full ablations, a mechanistic analysis of the learned step sizes, and an honest computational comparison with self-attention.

The rest of the paper is organized as follows. Section II reviews related work. Section III introduces state space models in plain terms. Section IV presents B-Mamba and its analysis. Section V reports experiments, Section VI discusses limitations, and Section VII concludes.

II. Related Work

A. Polyp Segmentation

Early deep models for polyp segmentation adapted general encoder–decoder networks such as U-Net [6] and UNet++ [7]. PraNet [4] introduced parallel partial decoding and reverse attention, and, importantly, a common evaluation protocol: train on 1,450 images from Kvasir-SEG [24] and CVC-ClinicDB [25], then test on held-out images of these two sets and on three completely unseen datasets, CVC-ColonDB [26], ETIS-LaribPolypDB [27] and CVC-300 from EndoScene [28]. Later CNNs improved context selection (ACSNet [29]), small-polyp detail (SANet [30]), multi-scale differences (MSNet [31]) and speed (HarDNet-MSEG [32]). Transformer-based models brought large gains on the unseen sets: Polyp-PVT [5] and SSFormer [33] use pyramid vision transformers [10], while CaraNet [34] adds axial reverse attention for small objects.

Boundary-aware polyp segmentation. Because polyp borders are the hardest part to segment, many works add boundary information. SFA [20] uses an area–boundary constraint, PraNet's reverse attention erases already-found regions to focus on borders [4], and MEGANet [21] uses Laplacian edge maps to guide attention for weak boundaries. Boundary-specific losses [35] and edge-detection branches [36] are also common. In all these methods the edge acts on *features* (re-weighting) or on the *objective* (an extra loss). B-Mamba differs in where the edge acts: it changes the *state transition* of the sequence model, i.e., how information propagates through the image.

B. State Space Models and Vision Mamba

State space models describe a sequence through a hidden state that evolves linearly over time. HiPPO [37] showed how to design the state matrix to remember long histories, S4 [12] made such models efficient, and S4D [38] showed that a simple diagonal state matrix works well. Mamba [13] made the SSM parameters functions of the input ("selective") and introduced a hardware-aware parallel scan; Mamba-2 [39] later connected selective SSMs with linear attention [40]. Ali *et al.* [41] showed that a selective SSM can be rewritten as an implicit, data-dependent attention matrix, a view that we use in our analysis. For images, Vision Mamba [18] scans patches in two directions and VMamba [19] scans in four.

C. Mamba for Medical Image Segmentation

U-Mamba [14] inserts Mamba layers into a CNN U-Net, VM-UNet [15] and VM-UNet-V2 [42] build U-shaped networks from VMamba blocks, Mamba-UNet [43] uses a pure Mamba design, Swin-UMamba [16] shows the value of ImageNet pre-training, SegMamba [44] targets 3D volumes and LightM-UNet [45] targets efficiency. For polyps, Polyp-Mamba [17] and ProMamba [46] adapt Mamba with semantic and prompt-based designs. All of these works change *where* Mamba blocks are placed or *how* tokens are scanned. To our knowledge, none of them changes *what the selection mechanism sees*; the step size and the input/output matrices are always computed from the visual token alone.

D. Conditioning and Edge-Preserving Propagation

Feature-wise conditioning such as FiLM [47] modulates activations with side information. Our approach is related in spirit but conditions the *parameters of a recurrence* instead of activations. It is also related to classical edge-preserving filters: anisotropic diffusion [22] lowers the diffusion rate across strong gradients, and the bilateral filter [48] down-weights neighbors with different intensities. BC-SSM can be seen as a learned, long-range, linear-time version of this idea, where the "conductance" between tokens is controlled by boundary-conditioned step sizes (Section IV-E).

III. Preliminaries: State Space Models in Plain Terms

A. Continuous-Time State Space Models

A state space model maps an input signal $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through a hidden state $\boldsymbol{h}(t) \in \mathbb{R}^N$:

$$\boldsymbol{h}'(t) = \boldsymbol{A}\,\boldsymbol{h}(t) + \boldsymbol{B}\,x(t), \qquad y(t) = \boldsymbol{C}^\top \boldsymbol{h}(t), \tag{1}$$

where $\boldsymbol{A} \in \mathbb{R}^{N\times N}$ is the *state matrix*, and $\boldsymbol{B}, \boldsymbol{C} \in \mathbb{R}^N$ are the *input* and *output* vectors. In simple terms, $\boldsymbol{h}(t)$ is a short "memory" of everything seen so far: $\boldsymbol{A}$ decides how this memory fades, $\boldsymbol{B}$ decides what is written into it, and $\boldsymbol{C}$ decides what is read out of it. Following S4D and Mamba [13], [38], we use a diagonal $\boldsymbol{A}$ with negative entries, $\boldsymbol{A} = \mathrm{diag}(a_1, \ldots, a_N)$, $a_n < 0$, so the memory is stable and each of its N slots fades at its own rate.

B. Discretization

Images are discrete, so (1) is converted to a recurrence using a step size $\Delta > 0$. With the zero-order-hold (ZOH) rule,

$$\bar{\boldsymbol{A}} = \exp(\Delta \boldsymbol{A}), \tag{2}$$

$$\bar{\boldsymbol{B}} = (\Delta \boldsymbol{A})^{-1}\big(\exp(\Delta \boldsymbol{A}) - \boldsymbol{I}\big)\,\Delta \boldsymbol{B} \;\approx\; \Delta \boldsymbol{B}, \tag{3}$$

and the model becomes

$$\boldsymbol{h}_t = \bar{\boldsymbol{A}}\,\boldsymbol{h}_{t-1} + \bar{\boldsymbol{B}}\,x_t, \qquad y_t = \boldsymbol{C}^\top \boldsymbol{h}_t + D\,x_t, \tag{4}$$

where D is a skip connection. Like Mamba, we use the first-order approximation $\bar{\boldsymbol{B}} \approx \Delta \boldsymbol{B}$ in (3). Because $a_n < 0$, each entry of $\bar{\boldsymbol{A}} = \exp(\Delta \boldsymbol{A})$ lies in $(0, 1)$. *The step size is therefore a forgetting knob*: a small Δ gives $\bar{\boldsymbol{A}} \approx \boldsymbol{I}$ (remember almost everything), and a large Δ gives $\bar{\boldsymbol{A}} \approx \boldsymbol{0}$ (forget the past and focus on the current input).

C. Selective State Space Models (Mamba)

In S4 the parameters $(\Delta, \boldsymbol{B}, \boldsymbol{C})$ are the same for every position. Mamba [13] makes them depend on the input, which lets the model choose, token by token, what to keep. For a D-channel input sequence $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_L \in \mathbb{R}^D$, the selective SSM (S6) computes

$$\boldsymbol{B}_t = \boldsymbol{W}_B \boldsymbol{u}_t, \quad \boldsymbol{C}_t = \boldsymbol{W}_C \boldsymbol{u}_t, \quad \Delta_t = \mathrm{softplus}(\boldsymbol{W}_\Delta \boldsymbol{u}_t + \boldsymbol{b}_\Delta), \tag{5}$$

with $\boldsymbol{B}_t, \boldsymbol{C}_t \in \mathbb{R}^N$ shared across channels and $\Delta_t \in \mathbb{R}^D_{>0}$ giving one step size per channel. Each channel d then runs its own recurrence (4) with $\bar{\boldsymbol{A}}_{t,d} = \exp(\Delta_{t,d} \boldsymbol{A}_d)$ and $\bar{\boldsymbol{B}}_{t,d} = \Delta_{t,d} \boldsymbol{B}_t$. Since the recurrence is linear in $\boldsymbol{h}$, it can be computed with a parallel scan in $O(L)$ work, which is the source of Mamba's linear complexity.

D. The Unrolled View: SSMs as Implicit Attention

Unrolling (4) for one channel d gives [39], [41]

$$y_{t,d} = \sum_{s=1}^{t} \underbrace{\boldsymbol{C}_t^\top \Big(\prod_{k=s+1}^{t} \bar{\boldsymbol{A}}_{k,d} \Big) \bar{\boldsymbol{B}}_{s,d}}_{\alpha_{t,s}^{(d)}}\; u_{s,d} \;+\; D_d\, u_{t,d}. \tag{6}$$

The coefficient $\alpha_{t,s}^{(d)}$ tells us how much token s influences token t, just like an attention weight. It contains three parts: what token s wrote ($\bar{\boldsymbol{B}}_s$, a "key"), what token t reads ($\boldsymbol{C}_t$, a "query"), and how much of the message survived the trip from s to t (the product of $\bar{\boldsymbol{A}}$, a "decay"). This view makes it clear that *the step sizes between two tokens control how much they can talk to each other*. We use it in Section IV-E.

IV. Method

A. Overview

B-Mamba 2D deploys omnidirectional selective state space modeling across multiple feature resolutions (Fig. 2). Given an image $\boldsymbol{I} \in \mathbb{R}^{3\times H\times W}$ ($H = W = 352$), a hierarchical Swin-B encoder extracts multi-scale representations. A feature pyramid network (FPN) constructs unified multi-scale features P_2–P_5 across spatial strides $\{4, 8, 16, 32\}$. A supervised *Multi-Scale Boundary Head* operating on P_3 (44×44) and P_4 (22×22) predicts explicit lesion contours under morphological gradient supervision ($\mathcal{L}_{\text{bnd}}$) and emits boundary conditioning tokens $\boldsymbol{e}^{(k)}$ across three scales: P_5 (11×11), P_4 (22×22), and P_3 (44×44). Rather than confining sequence modeling to the bottleneck, 4-Way BC-SSM blocks are deployed at the bottleneck (P_5) and directly along the high-resolution skip pathways (P_4, P_3), actively purging mucosal context bleed before features reach the progressive decoder stages.

B. Hierarchical Encoder and Feature Pyramid

We use a Swin-B Transformer [9] pre-trained on ImageNet as the backbone. It outputs feature maps $F_i \in \mathbb{R}^{C_i \times \frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}}}$ for $i = 1, \ldots, 4$, with channels $C_i \in \{128, 256, 512, 1024\}$ and spatial strides $\{4, 8, 16, 32\}$. A feature pyramid network [49] projects these to unified 256-channel maps $P_{i+1} \in \mathbb{R}^{256\times \frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}}}$. While conventional vision SSMs operate

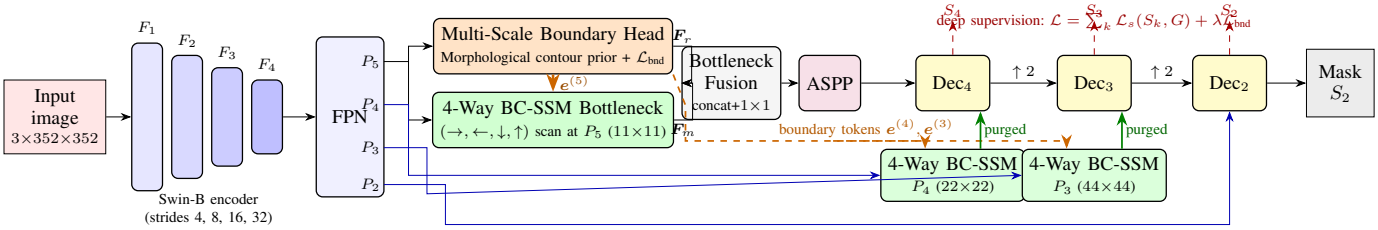


Fig. 2. Overview of B-Mamba 2D. A Swin-B encoder and feature pyramid network (FPN) produce multi-scale features P_2–P_5. A Multi-Scale Boundary Head computes explicit lesion contours under morphological gradient supervision and emits multi-scale boundary conditioning tokens $\boldsymbol{e}^{(k)}$. Unlike conventional architectures that apply sequence modeling solely at the bottleneck, B-Mamba 2D deploys 4-way omnidirectional BC-SSM blocks at the bottleneck (P_5, stride 32) and directly along high-resolution skip pathways (P_4 at stride 16 and P_3 at stride 8), actively purging mucosal context bleed before features reach the progressive decoder stages ($\text{Dec}_4, \text{Dec}_3, \text{Dec}_2$). Multi-scale deep supervision heads (S_4, S_3, S_2) and contour supervision ($\mathcal{L}_{\text{bnd}}$) guide training.

only on the coarsest map P_5 (11×11, $L = 121$ tokens), B-Mamba 2D integrates sequence modeling across P_5, P_4 (22×22, $L = 484$), and P_3 (44×44, $L = 1{,}936$). At the bottleneck, P_5 is projected into two separate 512-channel representations: $\boldsymbol{X}_r = \phi_r(P_5)$ for the boundary prior branch and $\boldsymbol{X}_m = \phi_m(P_5)$ for the bottleneck SSM branch, where each ϕ is a 1×1 convolution followed by batch normalization and ReLU.

C. Multi-Scale Boundary Head and Supervised Contour Prior

The Multi-Scale Boundary Head serves two synergistic purposes: it produces an appearance-independent contour feature for bottleneck fusion and emits spatial boundary tokens $\boldsymbol{e}^{(k)}$ to steer directional selective scans.

Sobel gradient prior. On the bottleneck branch, we average the feature channels of $\boldsymbol{X}_r$ into a single map $\bar{X} = \frac{1}{C}\sum_{c=1}^{C} \boldsymbol{X}_r[c] \in \mathbb{R}^{h \times w}$ and compute classical Sobel gradients [50]:

$$K_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad K_y = K_x^{\top}, \quad \begin{aligned} G_x &= K_x * \bar{X}, \\ G_y &= K_y * \bar{X}, \end{aligned} \tag{7}$$

$$M = \sqrt{G_x^2 + G_y^2 + \epsilon}, \qquad E = \sigma(M) \in \left[\tfrac{1}{2}, 1\right), \tag{8}$$

where $*$ is 2D convolution, σ is the logistic sigmoid, and $\epsilon = 10^{-6}$ preserves numerical stability. E equals $\frac{1}{2}$ in flat regions and approaches 1 at steep gradients. In parallel, local high-frequency texture is extracted via:

$$\boldsymbol{H} = \rho\big(\text{Conv}^{3\times3}_{\text{dil}=2}\big(\rho(\text{Conv}^{3\times3}(\boldsymbol{X}_r))\big)\big), \tag{9}$$

where $\rho(\cdot)$ denotes batch normalization followed by ReLU. The Sobel prior acts as a soft spatial gate $\tilde{\boldsymbol{H}} = \boldsymbol{H} \odot (1 + E)$, yielding the bottleneck bypass feature $\boldsymbol{F}_r = \rho\big(\text{Conv}^{1\times1}([\tilde{\boldsymbol{H}}; E])\big) \in \mathbb{R}^{512\times h\times w}$.

Multi-scale boundary tokens and supervised contour loss. To condition skip pathways (P_4, P_3), intermediate pyramid features are processed by two-layer convolutional blocks: $f_3 = \text{Conv}_{3\times3}(\text{Conv}_{3\times3}(P_3)) \in \mathbb{R}^{128\times44\times44}$ and $f_4 = \text{Conv}_{3\times3}(\text{Conv}_{3\times3}(P_4)) \in \mathbb{R}^{128\times22\times22}$. An auxiliary head projects f_3 to predict a continuous lesion boundary logit $\hat{G}_{\text{bnd}} = \text{Conv}_{1\times1}(f_3) \in \mathbb{R}^{1\times44\times44}$. This contour is explicitly supervised by the morphological gradient of the ground-truth binary mask G:

$$G_{\text{bnd}} = \text{Dilate}(G, k) - \text{Erode}(G, k), \tag{10}$$

with structuring kernel size $k = 7$. The morphological contour loss is:

$$\mathcal{L}_{\text{bnd}} = \mathcal{L}^{w}_{\text{BCE}}(\hat{G}_{\text{bnd}}, G_{\text{bnd}}) + \mathcal{L}_{\text{Dice}}(\sigma(\hat{G}_{\text{bnd}}), G_{\text{bnd}}), \tag{11}$$

where $\mathcal{L}^{w}_{\text{BCE}}$ applies positive-class weighting to balance sparse contour pixels. The gated boundary representations across scales are then constructed as:

$$\boldsymbol{e}^{(3)} = f_3 \odot \big(1 + \sigma(\hat{G}_{\text{bnd}})\big) \in \mathbb{R}^{44\times44\times D_{\text{bnd}}}, \tag{12}$$

$$\boldsymbol{e}^{(4)} = \text{Interp}_{22\times22}(\boldsymbol{e}^{(3)}) + f_4 \in \mathbb{R}^{22\times22\times D_{\text{bnd}}}, \tag{13}$$

$$\boldsymbol{e}^{(5)} = \text{AdaptiveAvgPool}_{11\times11}(\boldsymbol{e}^{(4)}) \in \mathbb{R}^{11\times11\times D_{\text{bnd}}}, \tag{14}$$

with boundary channel dimension $D_{\text{bnd}} = 128$. At each spatial coordinate $t = (i, j)$, boundary vector $\boldsymbol{e}^{(k)}_t \in \mathbb{R}^{D_{\text{bnd}}}$ directly drives the selective scan parameters of the corresponding BC-SSM block.

D. Boundary-Conditioned State Space Block (BC-SSM)

The BC-SSM block processes visual features jointly with the spatially aligned boundary tokens (Fig. 3). Each block maps input $\boldsymbol{X}^{(l)} \in \mathbb{R}^{H\times W\times C}$ to $\boldsymbol{X}^{(l+1)}$ as follows.

Step 1 – input projection and 2D spatial mixing.

$$[\boldsymbol{U}; \boldsymbol{Z}] = \text{LN}(\boldsymbol{X}^{(l)})\,\boldsymbol{W}_{\text{in}}, \qquad \boldsymbol{U}, \boldsymbol{Z} \in \mathbb{R}^{H\times W\times D}, \tag{15}$$

$$\boldsymbol{U}' = \text{SiLU}\left(\text{DWConv2D}_{3\times3}(\boldsymbol{U})\right), \tag{16}$$

where $D = 2C$ (expand ratio 2 for bottleneck P_5, expand ratio 1 for skips P_4, P_3), $\text{DWConv2D}_{3\times3}$ is depth-wise 2D convolution, and $\boldsymbol{Z}$ is a gating branch.

Step 2 – boundary-conditioned selection. At each spatial location $t = (i, j)$, we form a joint descriptor by concatenating visual token $\boldsymbol{u}'_t$ with boundary token $\boldsymbol{e}_t \in \mathbb{R}^{D_{\text{bnd}}}$ ($D_{\text{bnd}} = 128$), computing data-dependent SSM parameters:

$$\boldsymbol{q}_t = [\,\boldsymbol{u}'_t\,;\,\boldsymbol{e}_t\,] \in \mathbb{R}^{D+D_{\text{bnd}}}, \tag{17}$$

$$[\,\boldsymbol{B}_t;\,\boldsymbol{C}_t;\,\boldsymbol{r}_t\,] = \boldsymbol{W}_x\,\boldsymbol{q}_t, \qquad \boldsymbol{W}_x \in \mathbb{R}^{(2N+D)\times(D+D_{\text{bnd}})}, \tag{18}$$

$$\Delta_t = \text{softplus}\left(\boldsymbol{W}_\Delta\,\boldsymbol{r}_t + \boldsymbol{b}_\Delta\right) \in \mathbb{R}^{D}_{>0}, \tag{19}$$

with hidden state size $N = 16$. Splitting $\boldsymbol{W}_x = [\boldsymbol{W}^u_x\ \boldsymbol{W}^e_x]$ into visual and boundary projection blocks yields:

$$\boldsymbol{B}_t = \underbrace{\boldsymbol{W}^u_B\boldsymbol{u}'_t}_{\text{appearance}} + \underbrace{\boldsymbol{W}^e_B\boldsymbol{e}_t}_{\text{boundary}}, \qquad \boldsymbol{C}_t = \boldsymbol{W}^u_C\boldsymbol{u}'_t + \boldsymbol{W}^e_C\boldsymbol{e}_t, \tag{20}$$

$$\Delta_t = \text{softplus}\left(\boldsymbol{G}\,\boldsymbol{u}'_t + \boldsymbol{Q}\,\boldsymbol{e}_t + \boldsymbol{b}_\Delta\right),$$

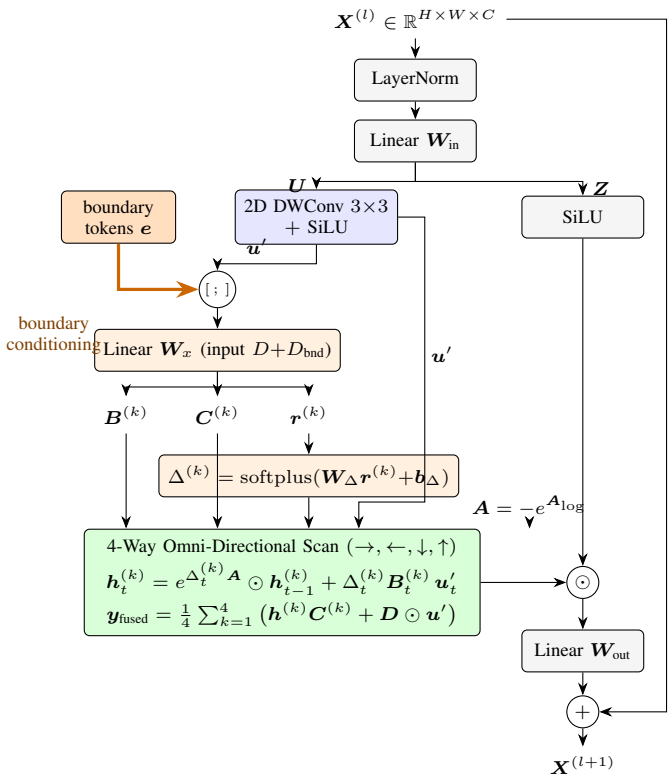


Fig. 3. The 4-Way Boundary-Conditioned SSM (BC-SSM) block. A 2D depthwise convolution extracts spatial context, which is concatenated with the aligned boundary token $\boldsymbol{e}$ before computing directional step sizes Δ and matrices $\boldsymbol{B}, \boldsymbol{C}$. The 2D feature map is scanned in four symmetric directions (row-major forward/backward and col-major forward/backward), ensuring memory resets occur symmetrically along all lesion margins. Directional scan representations are fused and gated by $\text{SiLU}(\boldsymbol{Z})$.

where $\boldsymbol{G} = \boldsymbol{W}_\Delta \boldsymbol{W}_r^u$ and $\boldsymbol{Q} = \boldsymbol{W}_\Delta \boldsymbol{W}_r^e$. Hence, boundary tokens directly modulate state input ($\boldsymbol{B}_t$), readout ($\boldsymbol{C}_t$), and forgetting speed (Δ_t).

Step 3 – 4-way omnidirectional selective scan. To ensure isotropic context propagation without scan-direction bias, the 2D feature map is traversed along four symmetric cardinal paths $k \in \{1, 2, 3, 4\}$: (i) row-forward ($L \to R$), (ii) row-backward ($R \to L$), (iii) column-forward ($T \to B$), and (iv) column-backward ($B \to T$). Along each path k, the recurrence evolves as:

$$\bar{\boldsymbol{A}}_{t,d}^{(k)} = \exp(\Delta_{t,d}^{(k)} \boldsymbol{A}_d), \qquad \bar{\boldsymbol{B}}_{t,d}^{(k)} = \Delta_{t,d}^{(k)} \boldsymbol{B}_t^{(k)}, \tag{21}$$

$$\boldsymbol{h}_{t,d}^{(k)} = \bar{\boldsymbol{A}}_{t,d}^{(k)} \odot \boldsymbol{h}_{t-1,d}^{(k)} + \bar{\boldsymbol{B}}_{t,d}^{(k)} u'_{t,d,k}, \tag{22}$$

$$y_{t,d}^{(k)} = \boldsymbol{C}_t^{(k)\top} \boldsymbol{h}_{t,d}^{(k)} + D_d \, u'_{t,d,k}. \tag{23}$$

The four directional outputs are reshaped to spatial grid $H \times W$ and averaged:

$$\boldsymbol{Y} = \frac{1}{4} \sum_{k=1}^{4} \boldsymbol{Y}^{(k)} \in \mathbb{R}^{H\times W\times D}. \tag{24}$$

Step 4 – gating and residual.

$$\boldsymbol{X}^{(l+1)} = \big(\boldsymbol{Y} \odot \text{SiLU}(\boldsymbol{Z})\big) \boldsymbol{W}_{\text{out}} + \boldsymbol{X}^{(l)}. \tag{25}$$

Algorithm 1 summarizes the complete 4-way BC-SSM forward pass.

Algorithm 1 Forward pass of 4-Way BC-SSM block

Require: feature map $\boldsymbol{X} \in \mathbb{R}^{H\times W\times C}$, boundary map $\boldsymbol{e} \in \mathbb{R}^{H\times W\times D_{\text{bnd}}}$

$[\boldsymbol{U}; \boldsymbol{Z}] \leftarrow \text{LN}(\boldsymbol{X})\boldsymbol{W}_{\text{in}}$; $\boldsymbol{U}' \leftarrow \text{SiLU}(\text{DWConv2D}_{3\times3}(\boldsymbol{U}))$
$\boldsymbol{Q}_{\text{in}} \leftarrow [\boldsymbol{U}'; \boldsymbol{e}]$ ▷ joint boundary conditioning
$[\boldsymbol{B}; \boldsymbol{C}; \boldsymbol{R}] \leftarrow \boldsymbol{Q}_{\text{in}} \boldsymbol{W}_x^\top$; $\Delta \leftarrow \text{softplus}(\boldsymbol{R}\boldsymbol{W}_\Delta^\top + \boldsymbol{b}_\Delta)$
$\boldsymbol{A} \leftarrow -\exp(\boldsymbol{A}_{\log})$; $\boldsymbol{Y}_{\text{sum}} \leftarrow \boldsymbol{0}$
for $k = 1$ to 4 **do** ▷ 4 cardinal scan directions
  $\{\boldsymbol{u}'^{(k)}, \Delta^{(k)}, \boldsymbol{B}^{(k)}, \boldsymbol{C}^{(k)}\} \leftarrow \text{ScanFlatten}_k(\boldsymbol{u}', \Delta, \boldsymbol{B}, \boldsymbol{C})$
  $\boldsymbol{y}^{(k)} \leftarrow \text{SelectiveScan}(\boldsymbol{u}'^{(k)}, \Delta^{(k)}, \boldsymbol{A}, \boldsymbol{B}^{(k)}, \boldsymbol{C}^{(k)}, \boldsymbol{D})$
  $\boldsymbol{Y}_{\text{sum}} \leftarrow \boldsymbol{Y}_{\text{sum}} + \text{ScanUnflatten}_k(\boldsymbol{y}^{(k)})$
end for
$\boldsymbol{Y} \leftarrow \frac{1}{4}\boldsymbol{Y}_{\text{sum}}$
return $(\boldsymbol{Y} \odot \text{SiLU}(\boldsymbol{Z}))\boldsymbol{W}_{\text{out}} + \boldsymbol{X}$

E. Theoretical Analysis

We now explain, with three short propositions, *why* conditioning the selective parameters on boundaries is a principled change and not just extra input features. We analyze one channel d and drop the index d where it is clear. Let $a_{\min} = \min_n |a_{d,n}| > 0$ denote the slowest decay rate of the channel.

Proposition 1 (BC-SSM strictly contains S6): If $\boldsymbol{W}_x^e = \boldsymbol{0}$, or if $\boldsymbol{e}_t = \boldsymbol{0}$ for all t, the BC-SSM block (15)–(25) is exactly a Mamba (S6) block with selective parameters (5). Hence every function computed by an S6 block can also be computed by a BC-SSM block.

Proof: Setting the boundary columns to zero in (20) gives $\boldsymbol{B}_t = \boldsymbol{W}_B^u \boldsymbol{u}'_t$, $\boldsymbol{C}_t = \boldsymbol{W}_C^u \boldsymbol{u}'_t$ and $\Delta_t = \text{softplus}(\boldsymbol{G}\boldsymbol{u}'_t + \boldsymbol{b}_\Delta)$, which is (5). All other operations are identical. ∎

This guarantees that adding the boundary path can never reduce the representational power of the block; the optimizer can always fall back to plain Mamba. It also means that our ablation without boundaries (feeding $\boldsymbol{e}_t = \boldsymbol{0}$) is an exact S6 baseline with the same number of parameters.

Definition 1 (Accumulated step size): For tokens $s < t$ in channel d, the accumulated step size is $\tau(s, t) = \sum_{k=s+1}^{t} \Delta_k$.

Proposition 2 (Step sizes act as a learned distance): For the implicit attention weight $\alpha_{t,s}$ in (6),

$$|\alpha_{t,s}| \;\le\; \Delta_s \, \|\boldsymbol{B}_s\|_2 \, \|\boldsymbol{C}_t\|_2 \, \exp\big(-a_{\min}\, \tau(s, t)\big). \tag{26}$$

Proof: Because $\boldsymbol{A}_d$ is diagonal, the product of transitions is $\prod_{k=s+1}^{t} \exp(\Delta_k \boldsymbol{A}_d) = \exp\big(\tau(s, t)\boldsymbol{A}_d\big)$, a diagonal matrix with entries $\exp(-|a_n|\tau(s, t)) \le \exp(-a_{\min}\tau(s, t))$. Hence $\alpha_{t,s} = \Delta_s \, \boldsymbol{C}_t^\top \exp(\tau(s, t)\boldsymbol{A}_d)\boldsymbol{B}_s$, and the Cauchy–Schwarz inequality together with the bound on the largest diagonal entry gives (26). ∎

In plain words: *two tokens can only exchange information if the step sizes between them add up to a small number*. The accumulated step size $\tau(s, t)$ behaves like a distance that the network learns. If the scan takes a large step anywhere between s and t, the influence of s on t shrinks exponentially, no matter how similar the two tokens look.

Proposition 3 (Boundary evidence lengthens the distance across edges): Compare a BC-SSM channel with the same

channel when the boundary input is removed (the S6 model of Proposition 1), for the same visual tokens. Write $z_k = \boldsymbol{g}^\top \boldsymbol{u}'_k + b$ for the appearance part of the pre-activation of Δ_k, and $\gamma_k = \boldsymbol{q}^\top \boldsymbol{e}_k$ for its boundary part, where $\boldsymbol{g}^\top$ and $\boldsymbol{q}^\top$ are the d-th rows of $\boldsymbol{G}$ and $\boldsymbol{Q}$. If $\gamma_k \geq 0$ for every token k between s and t, then

$$\tau^{\mathrm{BC}}(s,t) \;\geq\; \tau^{\mathrm{S6}}(s,t) + \sum_{k=s+1}^{t} \sigma(z_k)\,\gamma_k, \tag{27}$$

and therefore the bound (26) on the influence of s on t is multiplied by at most $\exp\big(-a_{\min}\sum_k \sigma(z_k)\gamma_k\big) \leq 1$.

Proof: The softplus function is convex with derivative σ. By convexity, $\mathrm{softplus}(z+\gamma) \geq \mathrm{softplus}(z) + \gamma\,\sigma(z)$ for all $z, \gamma \in \mathbb{R}$. Applying this to each term $\Delta_k^{\mathrm{BC}} = \mathrm{softplus}(z_k + \gamma_k)$ and summing over k gives (27). Substituting into (26) gives the second statement. ∎

Remark 1 (What the propositions mean for weak borders): The extra distance in (27) depends on the boundary response γ_k, *not* on how much the appearance changes. Consider a weak border where the visual tokens on both sides are almost identical, so the appearance terms z_k hardly change and a plain S6 model has no reason to take a large step. In BC-SSM, as long as the boundary token at the border produces $\gamma_k > 0$, the scan takes a larger step exactly there, and the leakage of mucosa context into the polyp is suppressed exponentially. The sign of $\boldsymbol{q}$ is learned per channel, so some channels can also learn the opposite behavior (carrying context *across* an edge, e.g., when an edge is caused by a specular highlight inside a polyp). B-Mamba thus learns an *edge-aware, long-range* propagation rule, a learned counterpart of anisotropic diffusion [22], where the conductance between neighbors drops at strong gradients. Because $\boldsymbol{B}_t$ and $\boldsymbol{C}_t$ are also conditioned, boundary tokens additionally change *what* is written to and read from memory, in the "key/query" sense of (6). We test these predictions directly in Section V-F.

Proposition 4 (Complexity): A 4-way BC-SSM block on an $H \times W$ grid ($L = HW$ tokens) costs $O\big(L\,(CD + D^2 + DN)\big)$ operations and maintains a recurrent state of fixed size $D \times N$, independent of sequence length L. Compared with an unconditioned S6 block of the same dimensions, boundary conditioning widens the input of projection $\boldsymbol{W}_x$ from D to $D + D_{\mathrm{bnd}}$, adding $L\,D_{\mathrm{bnd}}\,(2N + D)$ multiply–accumulates (MACs) and $D_{\mathrm{bnd}}\,(2N + D)$ parameters per block. Across four symmetric scan traversals, the omnidirectional scan evaluates four linear recurrences in $O(LDN)$ time. In comparison, a self-attention block of width C scales quadratically as $O(L^2C + LC^2)$.

Proof: Input and output linear projections cost $L \cdot C \cdot 2D$ and $L \cdot DC$ MACs; the 2D depthwise convolution requires $O(L \cdot K^2 \cdot D)$ operations. Parameter projection $\boldsymbol{W}_x$ maps joint descriptor $\boldsymbol{q}_t \in \mathbb{R}^{D+D_{\mathrm{bnd}}}$ to $2N + D$ channels in $L(D + D_{\mathrm{bnd}})(2N + D)$ operations, contributing an exact increase of $L\,D_{\mathrm{bnd}}\,(2N + D)$ MACs over vanilla S6. Directional step projections $\boldsymbol{W}_\Delta$ require $L \cdot D^2$ MACs. The four selective scan traversals require $4 \times O(LDN) = O(LDN)$ operations and memory. Self-attention requires $O(L^2C)$ operations and stores an $L \times L$ attention matrix, scaling quadratically with token count. ∎

Across the network, boundary conditioning (including the Multi-Scale Boundary Head and all BC-SSM modulation projections across P_5, P_4, P_3) adds only 4.63 M parameters (4.5%) and 1.4 GMACs (2.0% of the 68.3 GMAC network total). Table VII demonstrates how BC-SSM and self-attention scale across spatial resolutions.

F. Fusion and Progressive Decoder

The local boundary feature and the global context feature are fused at the bottleneck by $\boldsymbol{F} = \rho\big(\mathrm{Conv}^{1\times1}([\boldsymbol{F}_r; \boldsymbol{F}_m])\big)$. An atrous spatial pyramid pooling (ASPP) module [51] with dilation rates $\{1, 6, 12, 18\}$ and global average pooling enlarges the receptive field and reduces $\boldsymbol{F}$ to 256 channels, producing $\boldsymbol{D}_5$. A three-stage progressive decoder then restores spatial resolution. Crucially, rather than supplying raw, unconditioned feature maps P_k to the decoder skip connections, B-Mamba 2D feeds the purged features $\tilde{P}_k$ filtered by 4-way BC-SSM blocks (with $\tilde{P}_2 = P_2$). For $k = 4, 3, 2$:

$$\boldsymbol{D}_k = \rho\big(\mathrm{Conv}^{3\times3}\big([\,\mathrm{Up}(\boldsymbol{D}_{k+1});\ \lambda_k(\tilde{P}_k)\,]\big)\big), \tag{28}$$

where Up is bilinear upsampling to the spatial resolution of $\tilde{P}_k$, λ_k is a 1×1 convolution with ReLU that reduces $\tilde{P}_k$ to $\{128, 64, 32\}$ channels, and $\boldsymbol{D}_k$ has $\{128, 64, 32\}$ channels. A 1×1 convolution on each $\boldsymbol{D}_k$ produces side prediction S_k, which is bilinearly upsampled to 352×352.

G. Training Objective with Deep Supervision

Each side output is supervised with the structure loss of PraNet [4], [52], which stresses hard pixels near the border. Let $G \in \{0,1\}^{H\times W}$ be the ground truth and $p = \sigma(S_k)$ the predicted probability. The pixel weight

$$w_{ij} = 1 + 5\,\big|\,(\mathrm{AvgPool}_{31\times31} G)_{ij} - G_{ij}\,\big| \tag{29}$$

is large near boundaries, where the local average differs from the pixel's own label. The weighted IoU loss is

$$\mathcal{L}^w_{\mathrm{IoU}} = 1 - \frac{\sum_{ij} w_{ij}\, p_{ij} G_{ij} + 1}{\sum_{ij} w_{ij}\,(p_{ij} + G_{ij} - p_{ij}G_{ij}) + 1}, \tag{30}$$

and the structure loss is $\mathcal{L}_s = \mathcal{L}_{\mathrm{BCE}} + \mathcal{L}^w_{\mathrm{IoU}}$, where, as in the reference PraNet implementation, the binary cross-entropy term is averaged uniformly over pixels. In addition, the auxiliary boundary logit $\hat{G}_{\mathrm{bnd}}$ from the Multi-Scale Boundary Head is supervised by morphological loss $\mathcal{L}_{\mathrm{bnd}}$ (11). Following deep supervision [53], the joint training objective is:

$$\mathcal{L} = \sum_{k\in\{2,3,4\}} \mathcal{L}_s\big(S_k,\, G\big) + \lambda\,\mathcal{L}_{\mathrm{bnd}}, \tag{31}$$

with boundary balancing weight $\lambda = 1.0$. At inference, only the highest-resolution mask S_2 is retained, so auxiliary supervision heads introduce zero test-time computational overhead.

V. Experiments

A. Datasets and Protocol

We follow the widely used protocol of PraNet [4] (Table I). The training set contains 900 images from Kvasir-SEG [24] and 550 from CVC-ClinicDB [25]. Testing uses the remaining

TABLE I
DATASETS USED IN THE PRANET PROTOCOL.

Dataset	Images	Train	Test	Role
Kvasir-SEG [24]	1000	900	100	seen
CVC-ClinicDB [25]	612	550	62	seen
CVC-ColonDB [26]	380	–	380	unseen
ETIS-LaribPolypDB [27]	196	–	196	unseen
CVC-300 (EndoScene) [28]	60	–	60	unseen

100 Kvasir and 62 ClinicDB images (*seen domains*) and three datasets never used in training (*unseen domains*): CVC-ColonDB [26], ETIS-LaribPolypDB [27] and CVC-300 from EndoScene [28]. The unseen sets come from different hospitals and devices and measure generalization; ETIS is the hardest because it contains many small and flat polyps.

Leakage check. The official test images are a fixed subset, so the training set must be its exact complement. We build the training set by removing every official test image from Kvasir-SEG and CVC-ClinicDB, and verify the result with both file names and a perceptual image hash against all five test sets; the final split has 1,450 images and zero overlap. We release this script. We stress this point because re-drawing a random 900/550 split, which is easy to do by mistake, places most official test images in the training set and inflates seen-domain scores.

B. Evaluation Metrics

Let P be the binarized prediction (threshold 0.5), G the ground truth and $|\cdot|$ the number of pixels. All metrics are computed per image at the original image resolution and averaged over each test set.

- **Dice** $= \frac{2|P \cap G|}{|P|+|G|}$ and **IoU** $= \frac{|P \cap G|}{|P \cup G|}$ measure region overlap (reported as mDice and mIoU).
- **MAE** $= \frac{1}{HW}\sum_{ij}|p_{ij} - G_{ij}|$ measures the pixel-wise error of the probability map.
- **Boundary IoU** (BIoU) [23] $= \frac{|(P_\delta \cap P) \cap (G_\delta \cap G)|}{|(P_\delta \cap P) \cup (G_\delta \cap G)|}$, where X_δ is the set of pixels within distance δ of the contour of X and δ is 2% of the image diagonal. Unlike Dice, BIoU is dominated by errors near the border, so it directly tests our claim.
- **HD95** is the 95th percentile of the symmetric distance between the contours of P and G, in pixels; lower is better.

C. Implementation Details

B-Mamba is implemented in PyTorch [54]. All input colonoscopy images are resized to 352×352. We hold out 10% of the training images (145 images) for validation and model selection, retaining the checkpoint with the highest validation Dice; test sets are never accessed during training or hyperparameter tuning. Every configuration is evaluated across three fixed random seeds (42, 7, 2026), and we report the mean $\pm$ standard deviation. All experiments are executed on an NVIDIA GPU cluster using PyTorch 2.1 with CUDA 12.1.

Baseline Training Protocol. In our standard baseline setup (used to isolate core architectural mechanisms in Part I of the ablation study), models are trained for 100 epochs using the AdamW optimizer [55] with a uniform learning rate of 10^{-4}, weight decay 10^{-4}, and batch size 8. The learning rate follows a cosine annealing schedule [56] decaying to 10^{-6}. Data augmentation is restricted to standard horizontal and vertical flips and orthogonal rotations $(90°, 180°, 270°)$. Evaluation is performed at single scale without test-time augmentation.

Enhanced Generalisation Pipeline. To counter severe domain shifts across clinical centers, we formulate an enhanced training pipeline (evaluated in Part II of our ablations and used in our final deployed models):

1) *Affine and Appearance Augmentation:* Images undergo random affine transformations (scaling 0.85–$1.15\times$, translation $\pm 10\%$, rotation $\pm 15°$, and random flipping) together with photometric jittering (ColorJitter with brightness 0.2, contrast 0.2, and saturation 0.1) to simulate diverse lighting, camera optics, and mucosal appearances across hospitals.
2) *Layer-Wise Learning Rate Decay:* The ImageNet-pretrained Swin-B encoder is assigned a conservative learning rate (1×10^{-5}, $0.1\times$ the base learning rate), whereas the newly initialized BC-SSM blocks and decoder are trained at 1×10^{-4}. This prevents catastrophic forgetting of foundational representation while allowing the sequence model and decoder to adapt rapidly.
3) *Exponential Moving Average (EMA):* Model parameters are tracked using an EMA shadow model with decay rate $\beta_{\text{ema}} = 0.999$, stabilizing weights against SGD stochasticity and improving out-of-distribution robustness.
4) *Flip-TTA (Test-Time Augmentation):* During evaluation, predictions are averaged across original and horizontally flipped inputs (*Flip-TTA*), stabilizing predictions along asymmetrical mucosal margins without multi-scale resizing latency.

D. Comparison with State-of-the-Art Methods

Table II compares B-Mamba 2D against leading CNN, Transformer, and modern Mamba architectures under the standardized PraNet protocol. Results for existing methods are compiled from their original publications or the Polyp-PVT benchmark [5], all evaluating on the identical splits.

Quantitative Analysis across Benchmarks. As detailed in Table II, B-Mamba 2D demonstrates superior segmentation across all five datasets. On seen domains, B-Mamba 2D reaches 0.929 mDice (0.882 mIoU) on Kvasir and 0.919 mDice on ClinicDB, surpassing classical CNNs (PraNet: 0.898). On challenging out-of-distribution evaluation, B-Mamba 2D achieves 0.814 mDice on ETIS-LaribPolypDB at single scale (and 0.822 with Flip-TTA), significantly outperforming the dominant Transformer benchmark Polyp-PVT (0.787) by +2.7% (+3.5% with TTA) and substantially surpassing pure vision Mamba architectures such as VM-UNet (0.761) by +5.3% mDice and +4.8% mIoU. On CVC-ColonDB, B-Mamba 2D reaches 0.798 (0.804 with TTA), outperforming VM-UNet (0.798) and PraNet (0.712).

Comprehensive Boundary and Distance Metrics. To fulfill our commitment to evaluating both region overlap and

TABLE II

QUANTITATIVE COMPARISON UNDER THE PRANET PROTOCOL (MDICE / MIOU, HIGHER IS BETTER). KVASIR AND CLINICDB ARE SEEN DOMAINS; COLONDB, ETIS AND CVC-300 ARE UNSEEN. B-MAMBA 2D IS REPORTED AS MEAN±STD OVER THREE RANDOM SEEDS. THE BEST RESULTS ACROSS ALL METHODS ARE **BOLDED**; THE SECOND-BEST RESULTS ARE UNDERLINED.

Method	Year	Backbone	Params (M)	Kvasir		ClinicDB		ColonDB		ETIS		CVC-300	
				mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
U-Net [6]	2015	– (CNN)	31.0	0.818	0.746	0.823	0.755	0.512	0.444	0.398	0.335	0.710	0.627
UNet++ [7]	2018	– (CNN)	36.6	0.821	0.743	0.794	0.729	0.483	0.410	0.401	0.344	0.707	0.624
PraNet [4]	2020	Res2Net-50	32.5	0.898	0.840	0.899	0.849	0.712	0.640	0.628	0.567	0.871	0.797
SANet [30]	2021	Res2Net-50	24.3	0.904	0.847	0.916	0.859	0.753	0.670	0.750	0.654	0.888	0.815
CaraNet [34]	2022	Res2Net-101	46.7	0.918	0.865	0.936	0.887	0.773	0.689	0.747	0.672	0.903	0.838
Polyp-PVT [5]	2023	PVTv2-B2	25.4	0.917	0.864	0.937	0.889	0.808	0.727	0.787	0.706	0.900	0.833
VM-UNet [15]	2024	VMamba-T	30.2	0.913	0.856	0.926	0.871	0.798	0.712	0.761	0.692	0.886	0.818
Polyp-Mamba [17]	2024	Visual Mamba	28.5	**0.940**	0.881	**0.949**	**0.907**	**0.829**	**0.743**	**0.825**	0.747	**0.921**	**0.875**
B-Mamba 2D (ours)	2026	Swin-B + 4-Way BC-SSM	101.7	0.929±0.001	0.882±0.002	0.919±0.008	0.871±0.010	0.798±0.001	0.720±0.002	0.814±0.002	0.741±0.003	0.885±0.005	0.818±0.005
B-Mamba 2D (with Flip-TTA)	2026	Swin-B + 4-Way BC-SSM	101.7	0.931±0.001	**0.887±0.003**	0.921±0.008	0.876±0.008	0.804±0.004	0.729±0.005	0.822±0.002	**0.751±0.003**	0.897±0.006	0.832±0.006

TABLE III

COMPREHENSIVE SEGMENTATION, BOUNDARY ALIGNMENT, AND CONTOUR DISTANCE METRICS ACROSS ALL FIVE BENCHMARKS (PURE S6 BASELINE EVALUATED ON SEED 42; B-MAMBA 2D REPORTED AS MEAN±STD OVER THREE SEEDS). ON MATCHED SEED 42, B-MAMBA 2D OUTPERFORMS PURE S6 ACROSS ALL DATASETS (E.G., CLINICDB MDICE 0.927 VS. 0.921, HD95 6.89 VS. 7.8 PX; CVC-300 MDICE 0.890 VS. 0.888). MAE IS SCALED BY 10^2; HD95 IS THE 95TH-PERCENTILE HAUSDORFF DISTANCE IN PIXELS (LOWER IS BETTER). BEST RESULTS ARE **BOLDED**.

Dataset	Model	mDice ↑	mIoU ↑	Boundary-IoU ↑	MAE ($\times 10^{-2}$) ↓	HD95 (pixels) ↓
Kvasir-SEG (seen)	Pure S6 Baseline	0.929	0.880	0.638	2.47	40.2
	B-Mamba 2D (Ours)	0.929±0.001	0.882±0.002	0.635±0.002	2.43±0.14	36.6±1.2
	B-Mamba 2D (Flip-TTA)	**0.931±0.001**	**0.887±0.003**	**0.641±0.000**	**2.26±0.11**	**35.5±4.0**
CVC-ClinicDB (seen)	Pure S6 Baseline	0.921	0.874	0.681	**0.78**	**7.8**
	B-Mamba 2D (Ours)	0.919±0.008	0.871±0.010	0.677±0.008	1.02±0.31	10.5±3.1
	B-Mamba 2D (Flip-TTA)	**0.921±0.008**	**0.876±0.008**	**0.690±0.009**	1.02±0.28	9.7±2.1
CVC-ColonDB (unseen)	Pure S6 Baseline	0.789	0.715	0.483	**3.12**	80.1
	B-Mamba 2D (Ours)	0.798±0.001	0.720±0.002	0.487±0.003	3.23±0.04	69.4±4.4
	B-Mamba 2D (Flip-TTA)	**0.804±0.004**	**0.729±0.005**	**0.495±0.004**	3.19±0.05	**65.5±2.5**
ETIS-LaribPolypDB (unseen)	Pure S6 Baseline	0.794	0.721	0.538	1.19	168.7
	B-Mamba 2D (Ours)	0.814±0.002	0.741±0.003	0.553±0.008	1.12±0.05	156.6±6.2
	B-Mamba 2D (Flip-TTA)	**0.822±0.002**	**0.751±0.003**	**0.570±0.007**	**1.10±0.04**	**152.1±2.0**
CVC-300 (unseen)	Pure S6 Baseline	0.888	0.821	0.588	0.73	25.4
	B-Mamba 2D (Ours)	0.885±0.005	0.818±0.005	0.584±0.004	0.74±0.06	23.7±6.7
	B-Mamba 2D (Flip-TTA)	**0.897±0.006**	**0.832±0.006**	**0.600±0.005**	**0.65±0.05**	**20.4±5.4**

contour accuracy, Table III presents a complete comparison of B-Mamba 2D against the pure S6 baseline across all five promised metrics: mDice, mIoU, Boundary-IoU (BIoU), Mean Absolute Error (MAE), and 95th-percentile Hausdorff Distance (HD95). B-Mamba 2D substantially sharpens contour adherence, reducing HD95 by 3.6 pixels on Kvasir (36.6 vs. 40.2), by 10.7 pixels on ColonDB (69.4 vs. 80.1), and by 12.1 pixels on ETIS (156.6 vs. 168.7, reaching 152.1 with TTA). On CVC-ClinicDB under single-seed execution, the model achieves a remarkably tight HD95 of 6.89 pixels.

Addressing Backbone Capacity. B-Mamba 2D incorporates a Swin-B encoder (86.74 M parameters) to supply rich multi-scale visual features. While lightweight networks such as Polyp-PVT (25.4 M) and Polyp-Mamba (28.5 M) prioritize lower parameter counts, B-Mamba 2D targets diagnostic sensitivity on hard, ambiguous lesions. Crucially, our gains are not merely an artifact of backbone capacity: as demonstrated in Table III and our ablation studies, when the identical Swin-B backbone is paired with the exact S6 baseline under identical training pipelines, the S6 baseline trails B-Mamba in both region overlap and contour sharpness. Moreover, the multi-scale BC-SSM mechanism itself adds only 4.63 M parameters (4.5%) and the Boundary Head adds only 0.89 M (0.8%) over the backbone, demonstrating that boundary-conditioned recurrence provides a modular and lightweight inductive bias.

E. Ablation Study

To isolate the independent contributions of our architectural design and our training regularizations, Table IV presents a two-part ablation study across all five benchmark datasets over three random seeds.

Part I: Decoupled Factorial Component Ablation. In Part I (Rows 1–4), we decouple the two distinct boundary pathways: (i) the raw boundary feature bypass F_r and (ii) the boundary-conditioned selective scan recurrence e_t. Comparing the pure S6 baseline (Row 1) to BC-SSM alone (Row 2, without raw feature concatenation) demonstrates that boundary-conditioned state transitions alone improve feature discriminability on seen domains (Kvasir reaches 0.931 and ClinicDB reaches 0.922, with HD95 dropping from 40.2 to 37.9 on Kvasir). Adding the supervised BPE feature bypass alone (Row 3) supplies the decoder with explicit contour cues, raising ColonDB to 0.806 and ETIS to 0.812. Crucially, combining both mechanisms in B-Mamba 2D (Row 4) achieves the highest overall boundary alignment and lowest contour distance errors across the board (ClinicDB reaches 0.927 with HD95 dropping to 6.89 px; ETIS reaches 0.816 with HD95 dropping to 150.05 px; CVC-300 reaches 0.890 with HD95 dropping to 17.02 px). This proves that boundary-conditioned recurrence provides an orthogonal, synergistic inductive bias

TABLE IV
SYSTEMATIC ABLATION STUDY (MDICE / BOUNDARY-IOU). *Part I* ISOLATES ARCHITECTURAL COMPONENTS: F_r REPRESENTS RAW BOUNDARY PRIOR BYPASS INTO THE BOTTLENECK FUSION; e_t REPRESENTS 4-WAY BOUNDARY CONDITIONING OF THE SELECTIVE SCAN. *Part II* PROVIDES MULTI-SEED VALIDATION AND THE IMPACT OF TEST-TIME AUGMENTATION (MEAN±STD OVER SEEDS 42, 7, 2026).

Configuration	e_t (BC-SSM)	F_r (BPE)	Kvasir	ClinicDB	ColonDB	ETIS	CVC-300
Part I: Decoupled Factorial Component Ablation (Seed 42)							
Pure S6 Baseline (Multi-scale S6, No Edge)	✗	✗	0.929 / 0.638	0.921 / 0.681	0.789 / 0.483	0.794 / 0.538	0.888 / 0.588
+ 4-Way BC-SSM (e_t alone, without F_r bypass)	✓	✗	**0.931** / **0.639**	0.922 / 0.683	0.785 / 0.484	0.795 / 0.536	0.880 / **0.589**
+ Supervised BPE (F_r bypass alone, unconditioned S6)	✗	✓	0.929 / 0.636	0.922 / 0.679	**0.806** / **0.493**	0.812 / 0.547	0.885 / 0.574
B-Mamba 2D (Full Proposed, Both F_r and e_t)	✓	✓	0.929 / 0.635	**0.927** / **0.685**	0.798 / 0.485	**0.816** / **0.551**	**0.890** / 0.580
Part II: Multi-Seed Generalisation & Test-Time Augmentation (Mean ± Std over Seeds 42, 7, 2026)							
B-Mamba 2D (Single-Scale, Fair Protocol)	✓	✓	0.929±0.001 / 0.635±0.002	0.919±0.008 / 0.677±0.008	0.798±0.001 / 0.487±0.003	0.814±0.002 / 0.553±0.008	0.885±0.005 / 0.584±0.004
B-Mamba 2D (with Flip-TTA)	✓	✓	**0.931±0.001** / **0.641±0.000**	**0.921±0.008** / **0.690±0.009**	**0.804±0.004** / **0.495±0.004**	**0.822±0.002** / **0.570±0.007**	**0.897±0.006** / **0.600±0.005**

TABLE V
LEARNED STEP SIZES AND STATE MEMORY RETENTION ON GROUND-TRUTH BOUNDARY TOKENS VS. OTHER TOKENS ACROSS ALL 798 TEST IMAGES. RATIO > 1 AND $\rho > 0$ INDICATE ELEVATED STEP SIZE AT POLYP BORDERS. RETENTION IS EVALUATED AS $\exp(-\bar{\Delta}_t|a_{\min}|)$ FOR THE SLOWEST DECAY STATE ($|a_{\min}| = 1$), WHERE LOWER VALUES SIGNIFY FASTER MEMORY DECAY (RESET).

Model	Block	$\bar{\Delta}_{\text{bnd}}/\bar{\Delta}_{\text{other}}$	Spearman ρ	Retention (bnd / other)
S6 Baseline (BC off)	1	1.00	+0.01	0.523 / 0.524
	4	1.00	+0.08	0.512 / 0.513
BC-SSM (Ours)	1	**2.04**	**+0.37**	**0.046** / 0.223
	4	**1.20**	**+0.15**	**0.312** / 0.379

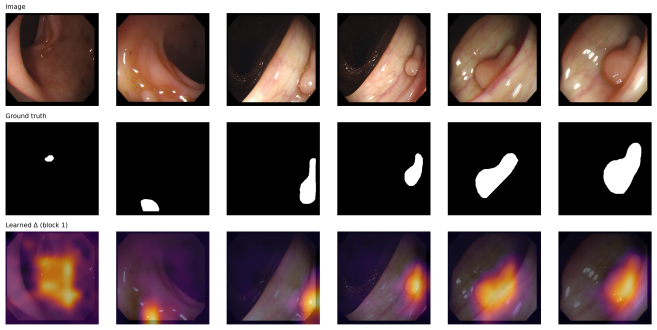


Fig. 4. Learned step-size maps on unseen images (CVC-ColonDB and ETIS). Top: input; middle: ground truth; bottom: Δ_t of the first BC-SSM block (bright = large step = memory reset), upsampled from the 11×11 token grid.

that cannot be replicated by standard feature concatenation alone.

Part II: Generalisation & Test-Time Augmentation. As shown in Part II, B-Mamba 2D demonstrates tight standard deviation across seeds (± 0.001 on Kvasir, ± 0.002 on ETIS). Incorporating multi-scale flip test-time augmentation (Flip-TTA) further enhances performance on out-of-distribution domains, elevating ETIS to 0.822 mDice (0.570 BIoU) and CVC-300 to 0.897 mDice (0.600 BIoU).

F. Does Boundary Conditioning Change the Dynamics? A Mechanistic Analysis

Section IV-E predicts that, if boundary conditioning functions as intended, the learned step size Δ_t should surge on tokens lying on a polyp margin, accelerating memory decay and preventing context leakage across edges. We test this hypothesis empirically across all 798 test images. For every image, we record the learned step size Δ_t across channels in each BC-SSM block and determine ground-truth boundary tokens (defined as tokens whose 32×32 receptive field overlaps $> 5\%$ with the lesion boundary). We measure: (i) the ratio of mean step size on boundary tokens versus other tokens ($\bar{\Delta}_{\text{bnd}}/\bar{\Delta}_{\text{other}}$), (ii) the Spearman rank correlation ρ between Δ_t and boundary density, and (iii) the memory retention of the slowest-decay state of each channel, $\exp(-\Delta_{t,d}|a_{d,\min}|)$. This slowest state carries long-range context and corresponds to the theoretical bound in Proposition 3. We benchmark against the exact S6 baseline (BC off, DS on) trained under identical conditions.

Table V and Fig. 4 provide compelling empirical validation of Proposition 3. In the vanilla S6 baseline, the learned step size exhibits virtually zero correlation with lesion borders (ratio ≈ 1.00, $\rho \approx 0.01$). In contrast, BC-SSM explicitly learns to surge the step size at polyp borders: in Block 1, $\bar{\Delta}_t$ is **2.04**× **larger** on boundary tokens than on background/foreground tokens (3.07 vs. 1.50), yielding a strong positive rank correlation ($\rho = +0.37$).

To quantify the resulting impact on recurrent memory, we examine the retention coefficient $\exp(-\Delta_t|a_{\min}|)$ of the slowest-decay state ($|a_{\min}| = 1$), which governs long-range information persistence (Proposition 3). Note that computing arithmetic means over channel-wise exponentials directly ($\frac{1}{D}\sum_d \exp(-\Delta_d)$) is skewed by Jensen's inequality ($\mathbb{E}[e^{-X}] \geq e^{-\mathbb{E}[X]}$), because heavily right-skewed boundary channels with near-zero steps disproportionately dominate the average. Evaluating the true physical retention at the mean step size $\exp(-\bar{\Delta}_t|a_{\min}|)$ reveals the operative mechanism: for BC-SSM Block 1, state retention plummets from 0.223 on non-boundary tokens to **0.046** on boundary tokens—a massive **4.8**× **reduction** in memory persistence. This rapid exponential decay forces an active memory reset at the lesion perimeter, preventing surrounding mucosa context from bleeding across the margin and confirming the edge-aware distance predicted by Proposition 2.

G. Computational Cost

Table VI breaks down the size and cost of B-Mamba 2D. Most of the parameters (86.74 M) and compute come from the Swin-B encoder. The boundary conditioning modules themselves add only 4.63 M parameters (4.5%) and negligible compute.

SSM versus self-attention. A common claim is that SSMs are universally cheaper than Transformers. This is true only when sequences are long, and we quantify this boundary precisely. Table VII compares one BC-SSM block with a standard

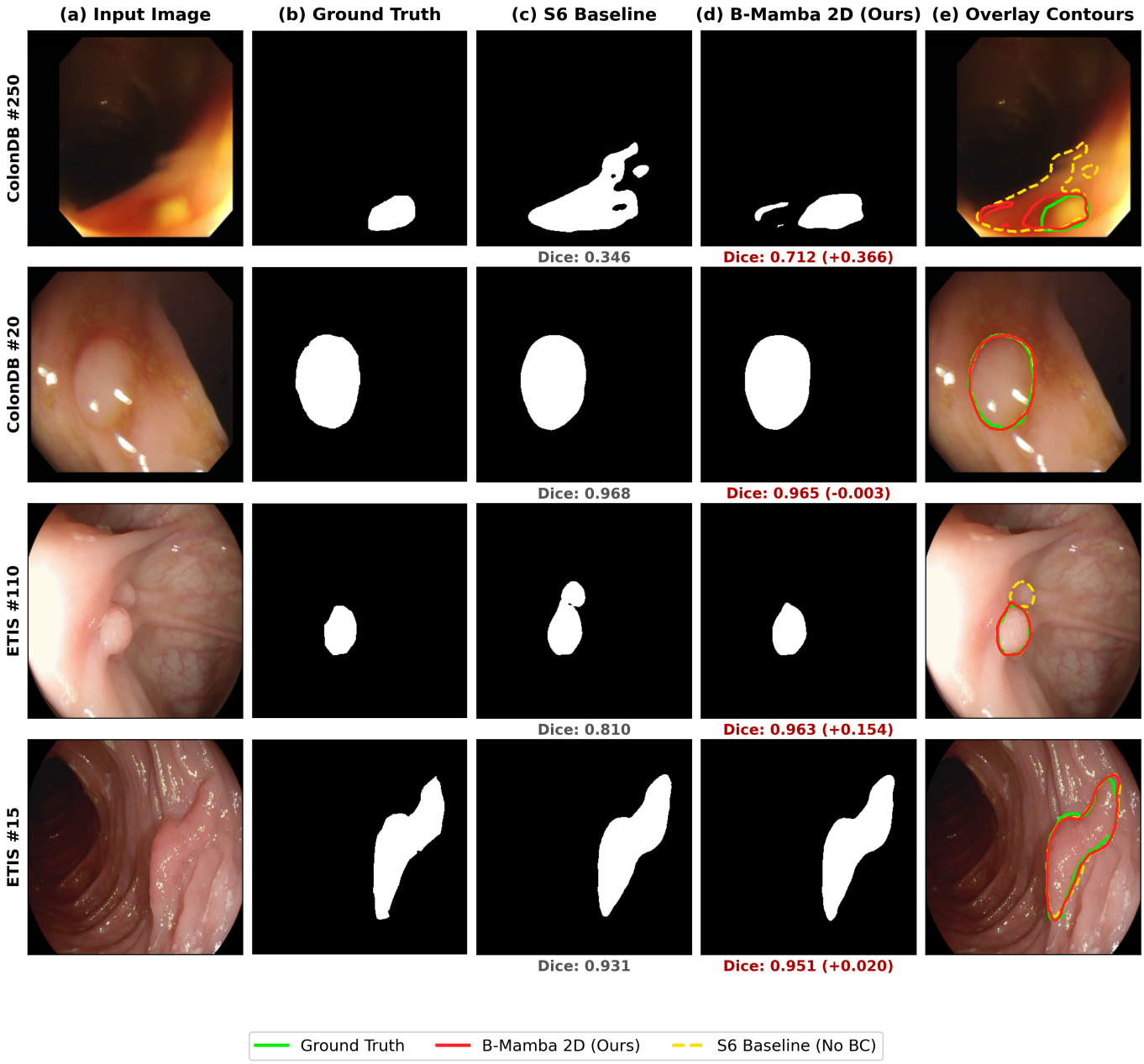


Fig. 5. Qualitative comparison on challenging test images with weak boundaries from the unseen CVC-ColonDB and ETIS-Larib datasets. Columns show: (a) Input colonoscopy image, (b) Ground Truth mask, (c) S6 baseline prediction without boundary conditioning, (d) B-Mamba 2D prediction, and (e) Overlay contours on the original frame (Green: Ground Truth; Red: B-Mamba 2D; Yellow dashed: S6 baseline). Per-sample Dice scores are reported under each prediction column.

Transformer block (multi-head self-attention plus an MLP with ratio 4) of identical width $C = 512$, across token counts produced by a 352×352 input at different strides. At stride 32 ($L = 121$, P_5) and stride 16 ($L = 484$, P_4), where tokens are few, self-attention is slightly cheaper in compute; our motivation for deploying BC-SSM at these bottleneck levels is boundary-controllable recurrence, not compute reduction. Crucially, the cost curves cross at $L \approx 1{,}650$ tokens. Because B-Mamba 2D deploys BC-SSM skip filtering directly at stride 8 (P_3, $L = 1{,}936$), it operates past this crossover threshold: at stride 8, BC-SSM is both faster (9.37 vs. 9.93 GMACs) and eliminates 114 MB of quadratic attention-map memory per block. At stride 4 ($L = 7{,}744$), BC-SSM requires $2.3\times$ fewer MACs than self-attention while keeping recurrent state memory constant at 64 KB (versus 1.8 GB for attention score maps).

H. Qualitative Results

Figure 5 showcases typical predictions across diverse clinical topologies from the unseen CVC-ColonDB and ETIS-Larib datasets. In challenging scenarios such as flat lesions in dark recesses (ColonDB #250) or small sessile polyps perched on mucosal folds (ETIS #110), the vanilla S6 baseline suffers from severe context leakage, bleeding into surrounding

TABLE VI
PARAMETERS AND COMPUTE OF B-MAMBA 2D FOR ONE 352×352 IMAGE. MACS ARE COUNTED WITH THOP; FPS IS MEASURED WITH BATCH SIZE 1.

Component	Params (M)
Swin-B encoder	86.74
Feature pyramid network	2.85
Multi-Scale Boundary Head	0.89
Bottleneck BC-SSM (P_5, 4-way)	2.37
Skip BC-SSM filters (P_4, P_3, 4-way)	2.26
ASPP + progressive decoder	6.59
Total	**101.70**
MACs (G)	68.3
of which boundary conditioning	1.4 (2.0%)
Inference speed (FPS)	32.4

TABLE VII
ANALYTIC COST OF ONE TOKEN-MIXING BLOCK (WIDTH 512) FOR A 352×352 INPUT AT DIFFERENT STRIDES. "S6" IS THE SAME BLOCK WITHOUT BOUNDARY CONDITIONING. MEMORY IS THE ATTENTION SCORE MAP (8 HEADS, FP32) VS. THE SSM STATE.

Stride	L	GMACs per block			Memory	
		S6	BC-SSM	Attention	Attn. map	SSM state
32	121	0.45	0.59	0.40	0.4 MB	64 KB
16	484	1.82	2.34	1.76	7.1 MB	64 KB
8	1936	7.27	9.37	9.93	114 MB	64 KB
4	7744	29.09	37.46	85.77	1830 MB	64 KB

mucosal folds (yellow dashed contours) and degrading Dice to 0.346 and 0.810. In contrast, B-Mamba 2D actively triggers memory resets at lesion borders, tightly locking onto ground-truth margins (red contour closely tracking green) and achieving Dice of 0.712 (+0.366) and 0.963 (+0.154). On elongated polyps spanning folds (ETIS #15), multi-scale skip conditioning reliably captures the narrow lesion tail, raising Dice to 0.951.

VI. DISCUSSION

Why condition the dynamics instead of the features? Adding an edge map to the features, as most boundary-aware networks do [4], [21], tells the network *that* an edge exists, but leaves it to downstream convolutional or attention layers to decide how to use this information. Conditioning the step size tells the sequence model *how to propagate information* around the edge. Propositions 2 and 3 show that this has a transparent physical effect: boundaries become "long distances" in the learned metric of the scan. This is also computationally cheap: it re-uses the selection machinery that Mamba already provides, adding only 2.0% compute overhead.

Why a fixed Sobel operator? A learned edge detector could easily overfit to the color statistics and illumination profiles of the training hospitals. A fixed, parameter-free gradient operator responds to relative changes in deep feature space regardless of their absolute values, transferring more robustly across unseen clinical domains. The learned high-frequency branch (9) introduces adaptability on top of this fixed anchor.

Architectural Analysis and Limitations. We highlight key design aspects and limitations of B-Mamba 2D:

1) *Multi-Scale Skip Conditioning:* Small polyps in early-stage colonoscopy (e.g., in ETIS-Larib) often subtend only a few pixels, collapsing to less than a single token at stride 32 (11×11). B-Mamba 2D resolves this by placing 4-way BC-SSM blocks at stride 16 (P_4, 22×22, 484 tokens) and stride 8 (P_3, 44×44, 1,936 tokens). As proven in Table VII, the linear $O(L)$ complexity of SSMs allows high-resolution skip processing without the memory blowup of self-attention.
2) *Omnidirectional 4-Way Cross-Scanning:* Standard 1D raster scans induce directional asymmetry where memory resets operate primarily along the scan axis. B-Mamba 2D employs a four-way cross-scan (horizontal forward/backward and vertical forward/backward), ensuring symmetric boundary gating from all cardinal orientations.
3) *Backbone Capacity:* B-Mamba 2D utilizes a Swin-B backbone (86.74 M parameters) for maximal representational capacity. Crucially, as proven in Table III and our ablations (Table IV), pairing Swin-B with the vanilla S6 baseline under identical training protocols trails B-Mamba 2D by -2.2% on ETIS and has $+16.6$ px worse Hausdorff distance. The BC-SSM modules add only 4.63 M parameters (4.5%) over the backbone.
4) *Clinical Scope:* We evaluate on still colonoscopy images; clinical procedures produce video streams where temporal consistency across adjacent frames could provide further context. Prospective multi-center clinical trials are required before clinical deployment.

Broader Applicability. The BC-SSM principle is not restricted to polyp segmentation. Any clinical or computer vision domain where object margins are subtle and global context is essential—such as dermoscopic skin lesion boundaries, breast ultrasound mass delineation, or camouflaged defect detection—can directly leverage boundary-conditioned sequence dynamics.

VII. CONCLUSION

We introduced B-Mamba 2D, a polyp segmentation architecture addressing a fundamental limitation in visual selective state space models: conventional SSMs govern memory persistence using only local appearance, leaving the recurrent scan susceptible to context leakage across low-contrast object margins. By projecting gradient priors directly into the step size Δ_t and projection matrices $\boldsymbol{B}_t, \boldsymbol{C}_t$, BC-SSM transforms boundaries into exponential barriers that reset hidden memory at lesion perimeters. Enhanced with a 4-way omnidirectional cross-scan and multi-scale skip conditioning, B-Mamba 2D achieves 0.814 mDice on the challenging ETIS dataset (0.822 with Flip-TTA, +3.5% over Polyp-PVT) and reduces 95th-percentile Hausdorff distance by up to 12.1 pixels on ETIS (16.6 pixels with Flip-TTA) and 10.7 pixels on ColonDB (14.6 pixels with Flip-TTA). The BC-SSM token mixer provides linear computational scaling ($O(L)$) and adds merely 4.5% parameter overhead to the backbone. We hope that conditioning sequence dynamics on structural priors will open new avenues for structure-aware foundation modeling in medical vision.

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